



Life-history and phenotypic traits of insectivorous songbirds breeding on reclaimed mine land reveal ecological constraints

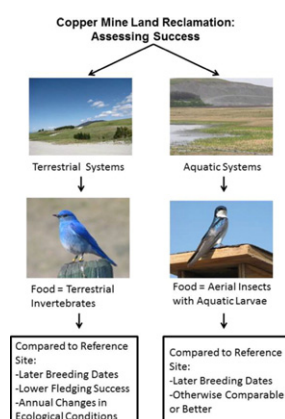
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HIGHLIGHTS

- Success of mine reclamation tested using reproductive success of insectivorous birds
- Two study species used terrestrial invertebrate food or insects with aquatic origins.
- Allows assessing success of reclamation of both terrestrial and aquatic systems
- Results show birds using aquatic food sources perform well, terrestrial food poor.
- Measures of individual performance are cost-effective tools for assessing reclamation.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 19 January 2016

Received in revised form 20 February 2016

Accepted 20 February 2016

Available online 27 February 2016

Editor: D. Barcelo

Keywords:

Mountain bluebird *Sialia currucoides*

Tree swallow *Tachycineta bicolor*

Ecological indicator

Impact assessment

Land reclamation

Food supplementation

ABSTRACT

Studies assessing impacts of industrial activities on wildlife typically examine population- or community-level responses. However, changes in measures such as species abundance or diversity are driven by cumulative responses of individuals to disturbance, and may take time to detect. Quantifying individual responses could allow us to foresee and mitigate future population declines resulting from industrial activities, while providing ecologically informative indices to assess quality of reclaimed land. We examined life-history and phenotypic traits of mountain bluebirds (*Sialia currucoides*) and tree swallows (*Tachycineta bicolor*) breeding on reclaimed copper mine lands in Canada over two years in comparison to a nearby undisturbed reference area. Bluebirds feed on terrestrial invertebrates, whereas swallows feed on adult forms of insects with aquatic larvae, allowing us to assess quality of both reclaimed terrestrial and aquatic systems as habitat for insectivorous birds. Supplemental feeding of bluebirds also was used to experimentally assess nutritional limitation of birds feeding on terrestrial invertebrates. Bluebirds on reclaimed land initiated clutches later, and in one year had lower fledging success compared to birds on the reference area. Tree swallows also bred later in the season on reclaimed land, but were otherwise comparable to or exceeded performance of birds on the reference area. Annual differences in responses of nestling bluebirds on the mine to supplemental feeding revealed an apparent switch in life-history strategy of parents between years, from brood reduction to brood survival, suggesting greater annual fluctuations in ecological conditions within terrestrial systems on reclaimed land. Sex differences in response of nestling bluebirds to food supplementation additionally suggested high within-brood competition for food on reclaimed land. We suggest that measures of avian life-history and phenotypic traits, particularly when assessed

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over multiple years using experimental approaches such as food supplementation, are informative and sensitive indices of the health of reclaimed terrestrial and aquatic systems.

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1. Introduction

Studies assessing ecological impacts of industrial activities on wild-life typically examine population- or community-level responses, such as species abundance or diversity. However, these measures are ultimately driven by the cumulative responses of individual organisms to environmental disturbance or degradation, and it may take time before effects at the population or community level are detected. For example, female northern goshawks (*Accipiter gentilis*) may skew brood sex ratios toward less-costly sons in food-limited habitats (daughters are larger, requiring more resources to produce; Ingraldi, 2005); the long-term population-level effect of this response, if sustained over multiple breeding seasons, may be a reduction in the number of breeding females, and hence population productivity, in future years. Anthropogenic noise has been associated with localized changes in age structure and pairing success of males (Habib et al., 2007), and with alterations in song (Francis et al., 2011), and reproductive output (Halfwerk et al., 2011) of several avian species. Birds may be more susceptible to parasites or pathogens in disturbed environments; for example, nests of tree swallows (*Tachycineta bicolor*) on reclaimed oil sands wetlands harbored more parasitic blow flies compared to nests on reference areas, and this was associated with reduced body condition of nestlings (Gentes et al., 2007). Individual responses such as these, if detected early, may allow us to foresee and mitigate future wildlife population declines resulting from industrial activities (Clutton-Brock and Sheldon, 2010).

Conducting individual-level studies may be deemed impractical for assessing land reclamation success, given the time and financial constraints associated with industrial environmental assessments. There is increasing evidence, however, that some measures of individual quality or condition are especially sensitive to environmental degradation, and so may respond quickly to changes in habitat quality. For example, the quality of condition-dependent traits which have evolved through sexual selection may decline rapidly under poor environmental conditions (Cockburn et al., 2008). Some of the best demonstrations of this are found in studies of plumage coloration in male birds, and these ornamental traits can therefore be important indicators of environmental health (Hill, 1995; Eeva et al., 2008). Identification and measurement of sensitive individual traits, however, has rarely been considered in the context of land reclamation or industrial impact assessment.

In this study, we investigated individual life-history and phenotypic traits of mountain bluebirds (*Sialia currucoides*) and tree swallows breeding in nest boxes on reclaimed mine tailings and waste rock within the Highland Valley Copper (HVC) operating area in British Columbia, Canada over two years. Mountain bluebirds feed primarily on insects gleaned from vegetation or the ground, whereas tree swallows are aerial insectivores, feeding predominantly on adult forms of invertebrates with aquatic larval stages (Power and Lombardo, 1996; McCarty, 1997; Winkler et al., 2011). By including both species in this study, we were able to assess the capacity of reclaimed tailings, waste rock and tailings ponds to support a variety of wildlife species, across a wider trophic range than would be possible by studying a single species. For comparison with birds on the HVC area, we also studied birds breeding on a nearby reference (REF) area, which has not historically been affected by mining operations. To experimentally investigate food availability as a specific factor affecting reproductive investment and offspring quality in mountain bluebirds on each study area, we additionally incorporated a food supplementation manipulation over both years. The goals of this study were to use measures of avian life-history and phenotypic traits to assess the effectiveness of existing mine reclamation protocols and to identify biologically relevant and informative indices of avian

reproductive investment and phenotypic traits that can be cost-effectively measured by industry to assess land reclamation success.

2. Methods

2.1. Study areas and field procedures

The Highland Valley Copper mine is a ~7000 ha open pit copper and molybdenum mining operation in south-central British Columbia, Canada. This study was conducted within the HVC Bethlehem Tailings and Reclaim area (elevation 1300–1550 m above sea level), which consists of reclaimed tailings and waste rock that have been enriched by application of biosolids and seeded with a grass-legume mix (Gardner, 2004). Patches of conifer forest are present in waste rock areas, and the margins of the site border on mature second growth mixed-wood forest. Several reclaimed tailings ponds and pit lakes are also present on the site (Hamaguchi et al., 2008). Mining on the study area occurred from the 1960s to the 1980s, and most of the land reclamation has been carried out since 1990 (Jones, 2001). The primary end land use goal for the site is cattle and wildlife grazing and forage production (Freberg and Gould Gizikoff, 1999).

Nest boxes were installed on fence posts throughout the HVC study area in 2001, and have been annually monitored to record nesting activities of mountain bluebirds (MOBL) and tree swallows (TRES). Reproductive data such as clutch size and overall nest success suggest that reproductive output for both species is comparable to that of populations breeding at similar elevations in undisturbed habitats (Howie, 2006; Highland Valley Copper, 2011). However, there is currently no comprehensive information on the reproductive investment or success of birds that are attracted to reclaimed lands on HVC, or how offspring quality (e.g. size at fledging) compares to that of birds breeding in areas not affected by mining operations.

The REF area consisted of established nest box trails located 10–30 km from the HVC site, at elevations of 1100–1400 m above sea level. Average ambient temperatures in nest boxes on the REF area in 2013 were approximately 1 °C higher throughout the breeding season compared to boxes on HVC, but minimum temperatures did not differ between areas (unpublished data). Nest boxes on the REF area were mounted on fence posts or snags in habitat types varying from clear-cuts, agricultural crop and grazing land as well as open grasslands bordered by mature mixed-wood or conifer forests. We monitored a total of 165 boxes on HVC and 194 boxes on the REF area during 2012 and 2013. On both study areas, boxes were typically mounted in pairs approximately 5 m apart, to allow both MOBL and TRES access to nest sites with minimal interspecific competition. Beginning in early May, we checked boxes every day to determine clutch initiation date and clutch size. Completed clutches were subsequently checked daily beginning 12 days after clutch completion to determine hatching date (designated nestling day 0) and hatching success of each egg in the clutch. Once all fertile eggs in a clutch had hatched, we weighed each nestling using an electronic balance (on day 1 or 2 for MOBL, day 2 for TRES, hereafter called hatching mass). When nestling MOBL were 15 days old and TRES were 16 days old, we measured mass (nearest 0.125 g, using a spring balance), and lengths of the longest primary wing feather (eighth in MOBL, ninth in TRES; nearest 0.5 mm with a ruler) and of the combined head and bill (as a measure of structural size; nearest 0.01 mm with digital calipers). We used a non-toxic pen to mark the tarsus of each nestling with a unique color combination when they were weighed after hatching, and refreshed marks throughout the nestling period so we were able to match each bird measured on day 15 or 16 to their hatching mass (see Section 2.3 (Data Analysis)). The phenotypic

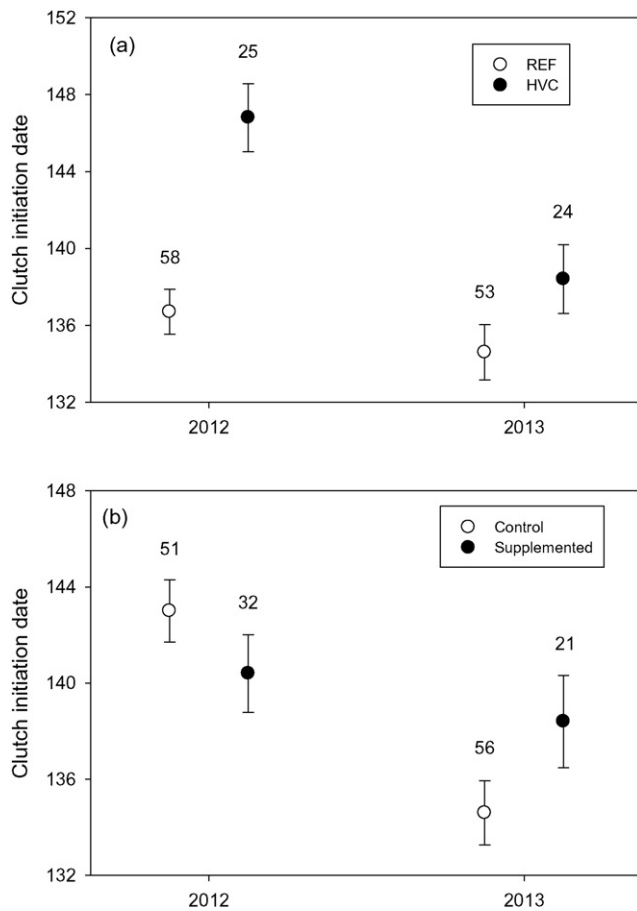


Fig. 1. Date of clutch initiation by female mountain bluebirds breeding over two years in relation to (a) whether they were breeding on reclaimed mine lands (HVC) or a nearby reference area (REF), and (b) whether they were provided with daily food supplements prior to breeding. Clutch initiation date is shown as ordinal date, where Jan 1 = 1, so larger values represent later initiation dates. Shown are estimated marginal means that account for effects of other predictors in the final models. Numbers above error bars refer to sample size (number of broods) for each group.

There were no differences in clutch size of MOBL between study areas or years; however, females that initiated breeding earlier in the season produced larger clutches, as did those that received supplemental food ($N = 50$, mean $\pm$ SE: 5.75 ± 0.09 ; controls: $N = 104$, mean $\pm$ SE: 5.47 ± 0.06 ; Table 1). Among 175 TRES clutches, there were similarly no differences in clutch size between areas or years; however, larger clutches were produced by ASY females and those that initiated breeding earlier in the season (Table 1).

Table 2

Results of binary logistic regression analyses that examined predictors of reproductive success in mountain bluebirds (MOBL) and tree swallows (TRES) breeding on reclaimed mine lands and a nearby reference area over two years. Parents of supplemented MOBL broods were provided with 12 g of mealworms and waxmoth larvae daily. We set 2013, Control Supplement and REF study area as reference categories for parameter estimates. CID refers to clutch initiation date, defined as the date the first egg in a clutch was laid.

Response variable	Predictor	MOBL			TRES		
		Wald	P	Estimate	Wald	P	Estimate
Hatching probability	Year	1.08	0.30		0.29	0.59	
	Area	0.009	0.93		4.34	0.04	-51.0 ± 24.5
	Supplement	2.12	0.15				
	CID	0.02	0.89		1.25	0.26	
	Area*CID				4.32	0.04	0.34 ± 0.2
Fledging probability	Year	4.30	0.04	-1.33 ± 0.6	7.11	0.008	-1.30 ± 0.5
	Area	4.84	0.03	-1.41 ± 0.6	6.03	0.01	20.3 ± 8.3
	Supplement	4.81	0.03	1.80 ± 0.8			
	CID	3.29	0.07	0.07 ± 0.04	0.001	0.99	
	Area*CID				6.20	0.01	-0.13 ± 0.05

Table 3

Results of general linear models assessing predictors of reproductive success in mountain bluebirds breeding on reclaimed mine lands and a nearby reference area over two years. Parents of supplemented broods were provided with 12 g of mealworms and waxmoth larvae daily. We set 2013, Control Supplement and REF study area as reference categories for parameter estimates. CID refers to clutch initiation date, defined as the date the first egg in a clutch was laid.

Response variable	Predictor	F	P	Estimate
Proportion hatched	Year	0.93	0.34	
	Area	1.30	0.26	
	Supplement	0.05	0.82	
	CID	0.009	0.93	
Proportion fledged	Year	5.64	0.02	-1.03 ± 0.4
	Area	1.38	0.24	
	Supplement	0.16	0.69	
	CID	0.04	0.84	
	Year*Area	6.25	0.01	0.19 ± 0.07
	Year*CID	5.15	0.03	0.01 ± 0.003

3.2. Reproductive success

We assessed whether or not at least one egg hatched for 148 MOBL clutches; of these, only 14 failed to hatch, and there were no differences in hatching success between study areas (Table 2). There was also no effect of food supplementation on probability of successfully hatching, and no differences in relation to year or timing of breeding (Table 2). Hatching probability for 200 TRES clutches varied with clutch initiation date, but this effect depended on study area (Table 2). Analyzing each study area separately revealed that hatching success did not vary with timing of breeding on the REF area ($P = 0.26$), but increased with later dates of clutch initiation on HVC (Wald = 3.69, $P = 0.05$, Estimate = 0.308 ± 0.2).

For 131 MOBL nests that hatched at least one young, the probability of fledging at least one young was higher on the REF area than HVC, and on both study areas was increased by supplemental feeding (Table 2). Fledging probability was also greater in 2013 compared to 2012, and showed a marginal increase with clutch initiation date (Table 2). Analysis of 185 nests of TRES that hatched at least one young showed similar annual variation in fledging probability, with higher success in 2013 than 2012, but we also detected an interaction between study area and clutch initiation date (Table 2). In contrast to the probability of hatching, when we separated the data by study area we found that the probability of TRES fledging young on reclaimed mine land decreased with timing of breeding (Wald = 8.35, $P = 0.004$, Estimate = -0.13 ± 0.05) but did not vary with clutch initiation date on the REF area ($P = 0.97$).

Among MOBL pairs that successfully hatched at least one egg ($N = 152$), the proportion of eggs in the clutch that hatched did not vary according to study area, supplementation, year or timing of breeding

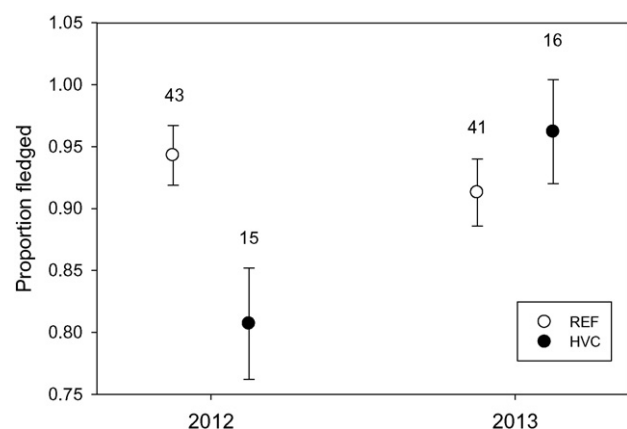


Fig. 2. Proportion of young fledging from broods of mountain bluebirds breeding on reclaimed mine lands (HVC) and a nearby reference area (REF) over two years. The analysis only included broods that successfully fledged at least one young. Shown are estimated marginal means that account for effects of other predictors in the final model. Numbers above error bars refer to sample size (number of broods) for each group.

(Table 3). Similarly, in TRES clutches ($N = 185$) the proportion hatching did not vary by area ($F = 1.84$, $P = 0.18$), year ($F = 0.09$, $P = 0.77$), or clutch initiation date ($F = 0.85$, $P = 0.36$). When we further focused only on nests that fledged at least one young, however, we found that the proportion of young fledging in MOBL broods ($N = 115$) was lower on reclaimed mine lands compared to the REF area in one of the two years (Table 3; Fig. 2). The proportion of young that fledged within broods of MOBL on both study areas also declined with clutch initiation date in 2013, but did not vary with timing of breeding in 2012 (Table 3). There was no additional effect of food supplementation on proportion of young that MOBL fledged on either study area (Table 3). Among TRES broods ($N = 153$), the proportion of young that fledged did not differ between study areas ($F = 0.57$, $P = 0.45$), year ($F = 2.48$, $P = 0.12$), female ages ($F = 0.47$, $P = 0.50$), or with clutch initiation date ($F = 0.54$, $P = 0.47$).

3.3. Offspring size at fledging

For body mass, eighth primary feather length, and head-bill length of 625 nestling MOBL prior to fledging, we detected interactions between study area and food supplementation that varied both by year and nestling sex (Table 4). There was no effect of supplementation on mass or feather length of MOBL on the REF area in either year, although nestlings in supplemented broods were structurally larger than controls;

however, on HVC, supplemental feeding of birds had opposing effects in each year (Fig. 3). In 2012, nestling MOBL in supplemented broods on HVC were structurally smaller, weighed less and developed shorter flight feathers than control broods on HVC and both treatment groups on the REF area. In contrast, supplementation in 2013 increased all nestling measures on HVC relative to control broods, and in fact allowed nestlings to attain body size and mass equivalent to that of both treatment groups on the REF area (Fig. 3). In addition, female nestlings raised in supplemented broods on HVC were lighter, smaller and had shorter flight feathers than females in control broods, whereas there were no effects of supplementation on male nestlings raised on HVC (Table 4; Fig. 4). In contrast, supplementation increased mass and head-bill length of male and female nestlings equally on the REF area (Fig. 4).

Body mass of nestling TRES ($N = 799$) decreased with hatching date more strongly on the HVC area, and more so in 2013 than 2012, although birds raised on HVC were overall heavier by day 16 than on the REF area (main effect of Area, Area*Hatch Date and Year*Hatch Date interactions; Table 5). Length of outer primary feathers at day 16 did not vary by hatching date, but increased more strongly with relative position in the brood size hierarchy for nestling TRES raised on REF than on HVC (Area*Hatch Mass; Table 5). Head-bill length of nestling TRES was marginally greater overall on HVC than REF, and declined with hatching date more strongly in 2013 (main effect of Area, Year*Hatch Date interaction; Table 5).

4. Discussion

We studied life-history and phenotypic traits of MOBL and TRES breeding in nest boxes on reclaimed mine tailings and waste rock areas over two years to assess the quality of reclaimed terrestrial and aquatic systems as habitat for insectivorous birds. This study allowed us to identify variables that are most sensitive to variation in environmental quality, and hence that may be valuable early indicators of land reclamation success or environmental disturbance. MOBL breeding on reclaimed mine lands initiated egg laying later in the season, and in one year were both less likely to fledge young once eggs had hatched, and fledged a smaller proportion of young in their broods compared to birds on the REF area. We did not detect any differences in clutch size or hatching success between study areas; in combination, these results suggest that costs associated with egg production and incubation are not differentially experienced by MOBL on reclaimed mine lands, whereas in some years, MOBL breeding on these reclaimed lands have more difficulty rearing young to independence. The magnitude of differences between study areas varied annually, with reduced productivity of MOBL on HVC more evident in 2012; this suggests greater annual

Table 4
Results of linear mixed models examining predictors of day 15 size and body mass of 625 nestling mountain bluebirds raised on reclaimed mine lands and a reference area (REF) over two years. Nestling sex was determined by plumage color, and hatching mass represents the standardized mass of each nestling at day 1 or 2 after hatching, relative to the mass of all young in the brood. Parents of supplemented broods were provided with 12 g of mealworms and waxmoth larvae daily. We set 2013, Control Supplement, females and REF area as reference categories for parameter estimates. Brood identity was fitted as a random effect to control for common brood environment and maternal effects among siblings, and predicted all phenotypic traits (Mass: Wald $Z = 6.36$; Primary: Wald $Z = 7.11$; Head-Bill: Wald $Z = 6.48$; all $P < 0.001$).

Predictor	Body mass (g)				8th primary feather length (mm)				Head-bill length (mm)			
	df	F	P	Estimate	df	F	P	Estimate	df	F	P	Estimate
Year	1119	2.01	0.16		1120	31.5	<0.001	-12.2 ± 2.3	1117	21.8	<0.001	-2.06 ± 0.4
Area	1119	1.56	0.21		1120	26.6	<0.001	-3.25 ± 1.9	1117	5.96	0.02	-0.25 ± 0.4
Sex	1495	17.1	<0.001	2.37 ± 0.49	1474	0.09	0.77		1487	0.10	0.75	
Supplement	1125	0.47	0.50		1128	0.84	0.36		1124	0.01	0.92	
Hatching mass	1465	25.6	<0.001	0.72 ± 0.17	1462	99.8	<0.001	1.68 ± 0.2	1462	68.5	<0.001	0.23 ± 0.03
Year*Area	1119	0.11	0.75		1120	1.29	0.26		1117	0.07	0.80	
Year*Supp	1125	4.93	0.03	4.21 ± 1.59	1128	11.2	0.001	12.2 ± 3.1	1124	13.5	<0.001	-2.52 ± 0.5
Area*Supp	1125	1.03	0.31		1128	4.18	0.04	1.62 ± 2.3	1124	6.38	0.01	-0.28 ± 0.4
Area*Sex	1495	9.28	0.002	2.24 ± 0.58	1474	3.16	0.08		1487	0.35	0.56	
Supp*Sex	1495	5.54	0.02	-1.97 ± 0.7	1474	2.65	0.10		1487	4.23	0.04	0.51 ± 0.2
Area*Hatch Mass	1465	5.48	0.02	0.46 ± 0.2	1462	4.32	0.04	0.50 ± 0.2				
Sex*Hatch Mass					1476	5.26	0.02	0.53 ± 0.2				
Year*Area*Supp	1125	5.39	0.02	4.31 ± 1.9	1128	13.0	<0.001	12.7 ± 3.5	1124	17.7	<0.001	2.68 ± 0.6
Area*Supp*Sex	1495	7.61	0.006	2.13 ± 0.77	1474	5.62	0.02	2.25 ± 1.0	1487	4.17	0.04	0.51 ± 0.2

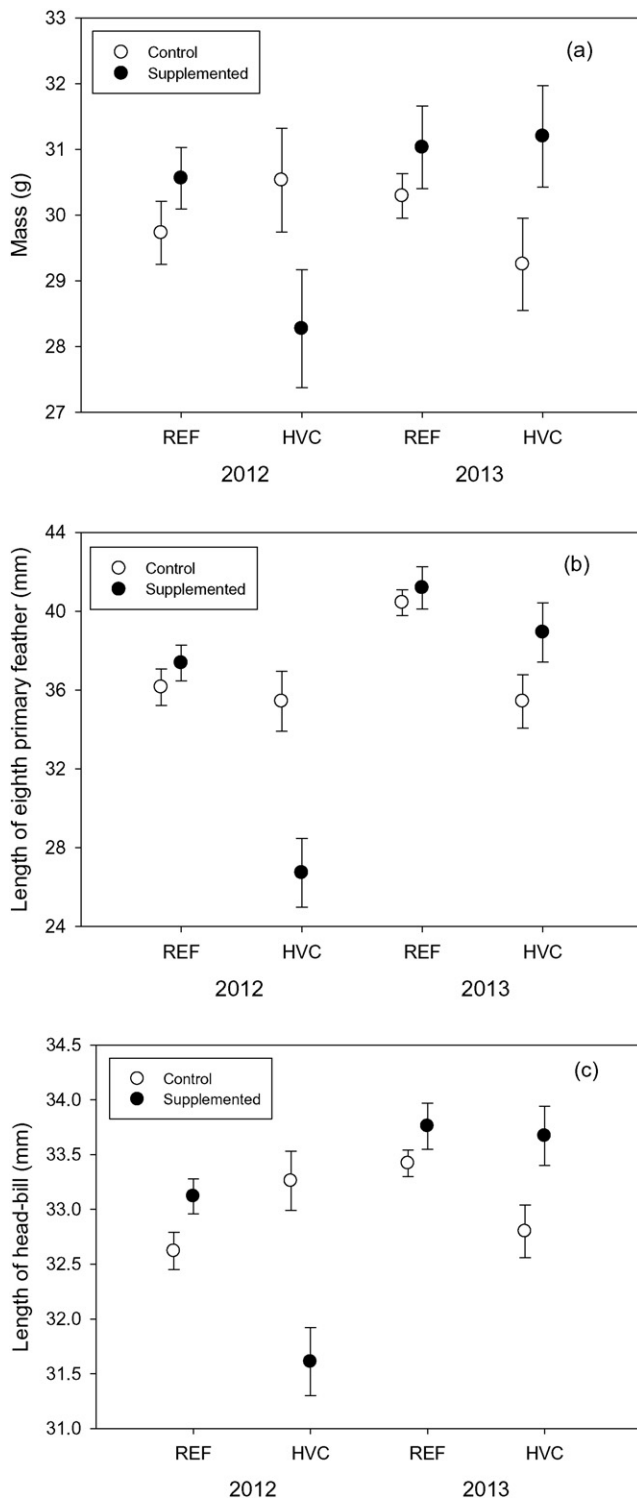


Fig. 3. Phenotypic traits of 15 day old nestling mountain bluebirds raised on reclaimed mine lands (HVC) and a nearby reference area (REF) over two years, shown in relation to whether or not the parents were provided with supplemental food throughout the nestling period. Linear mixed models included brood identity as a random effect to control for common brood environment and maternal effects among siblings. Values shown are estimated marginal means that account for effects of other predictors in final models.

fluctuations in ecological conditions such as terrestrial invertebrate availability on reclaimed mine lands compared to the REF area.

Clutch size is highly heritable in birds (e.g., McCleery et al., 2004) and so may not respond strongly to variation in habitat quality. Indeed, clutch size in MOBL, while showing a seasonal decline and response to

food supplementation on both study areas, showed no overall differences between study areas (Table 1). In contrast, avian life-history traits such as timing of breeding show considerable within-individual plasticity in response to annual environmental variation (Charmantier and

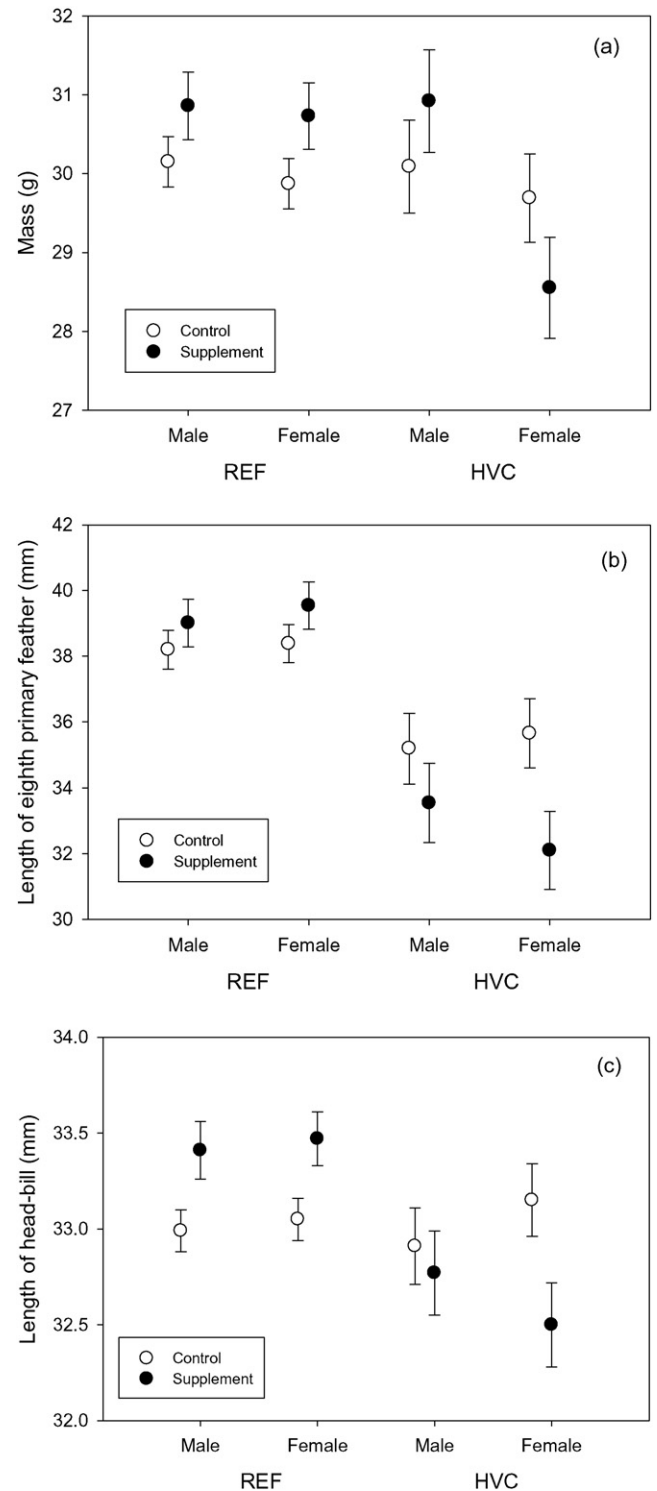


Fig. 4. Site-dependent sex differences in response to food supplementation, based on measurements of (a) body mass, (b) flight feather length and (c) head-bill length of 15 day old nestling mountain bluebirds raised on reclaimed mine lands (HVC) and a nearby reference area (REF). Parents of supplemented broods were provided with 12 g of mealworms and waxmoth larvae daily, throughout the nestling period. Linear mixed models included brood identity as a random effect to control for common brood environment and maternal effects among siblings. Values shown are estimated marginal means that account for effects of other predictors in the final models.

Gienapp, 2014), and we detected large differences between study areas in clutch initiation date for MOBL nests, particularly in 2012 (Fig. 1a). Hence, measuring avian clutch size alone will not allow industry to differentiate between high- and low-quality reclaimed terrestrial environments, and we suggest that the focus of reclamation assessment should instead be on clutch initiation date as a measure of female avian reproductive investment that represents a more sensitive indicator of environmental quality.

Similar to MOBL, TRES breeding on reclaimed mine lands initiated clutches later in the season. Earliest clutches of TRES on HVC were less likely to hatch; however, TRES breeding on HVC produced heavier and marginally larger young than birds breeding on the REF area, and fledging success did not differ between study areas in either year. Additionally, offspring phenotype was more strongly influenced by timing of breeding and position of individual nestlings within the brood size hierarchy at hatching on surrounding reference areas than on reclaimed mine lands (Table 5), suggesting more consistent food availability over the breeding season, and lower within-brood resource competition, for aerial insectivores on HVC. Aerial insectivores such as swallows typically feed on prey with an aquatic larval phase (McCarty, 1997); hence, these results indicate that conditions on reclaimed mine lands early in the season are generally less favourable for birds relying on aquatic invertebrate food resources, but resource availability for TRES is equivalent or greater on HVC compared to the REF area once birds reach the brood-rearing phase. Given that reclamation of aquatic ecosystems on HVC included the addition of aquatic invertebrates from nearby undisturbed systems (Hamaguchi et al., 2008), our results suggest that this approach has effectively restored aquatic systems to a level that can support breeding of aerial insectivores such as TRES.

According to life-history theory, one of the most common trade-offs for organisms during reproduction is between number and quality of young (Stearns, 1992). Under conditions of food limitation, parents may focus their investment on those young with the greatest fitness value, to the detriment of the remaining offspring (e.g., Howe, 1978). Between breeding attempts or seasons, songbirds may exhibit switches between brood reduction and brood survival strategies in response to variation in environmental conditions, particularly food availability (e.g., Du et al., 2014). Our supplemental feeding experiment revealed an apparent switch in life-history strategy of adult MOBL on reclaimed mine lands between study years, evidenced by changes in offspring phenotype in response to supplemental feeding of parents. In 2012, the year with later clutch initiation dates and overall reduced fledging success, the majority of offspring produced by supplemented parents on HVC exhibited reduced size and mass prior to fledging (Fig. 3). This suggests that MOBL on HVC followed a brood reduction strategy by concentrating their feeding efforts on only the largest or most vocal young in the brood, to the detriment of all other nestlings, or by simply switching to self-maintenance when given access to supplemental food. In contrast, in 2013, supplemental food appeared to have been distributed more equitably among siblings to promote brood survival, as supplemental feeding of parents on HVC allowed nestlings to

collectively achieve a mass and size at fledging comparable to all nestlings produced on the REF area, and which exceeded control broods on HVC (Fig. 3). Consequently, MOBL on HVC in 2013 were also able to fledge a greater proportion of young in their broods (Fig. 2). These results provide further evidence for greater annual fluctuations in ecological conditions such as natural availability of terrestrial invertebrate food sources on reclaimed mine lands compared to habitats on reference areas located at similar elevations and exposed to similar climatic conditions, and highlight that inter-annual unpredictability or variability may be an important metric by which to assess quality of reclaimed terrestrial systems as habitat for wildlife.

For all nestling MOBL measures, female young raised on reclaimed mine land showed a decline in performance when parents were provided with supplemental food, whereas both sexes consistently benefited from supplementation on the REF area (Fig. 4). Indeed, this differential response of male and female nestlings to supplementation on reclaimed land resulted in sexual dimorphism in body mass by day 15 that was not evident under control conditions or on the REF area (Fig. 4a). As with annual variation in food supplementation effects on offspring phenotype in MOBL on reclaimed mine land, this suggests greater within-brood competition for limited resources on HVC: parent MOBL may preferentially feed supplemental food to sons on HVC, either selectively, or simply because male nestlings are more competitive than female siblings (due in part to higher levels of circulating androgen hormones; von Engelhardt et al., 2006). This further suggests that parent birds experience stronger trade-offs between offspring number and quality on reclaimed mine land, and so tend to exhibit a brood reduction strategy of focusing investment on young with the greatest apparent fitness value (Howe, 1978; Stearns, 1992). In this context, such a strategy resulted in sexual dimorphism in offspring mass in response to experimental food supplementation (Fig. 4a). We therefore suggest that sex-specific responses to parental supplementation, and nestling growth responses to supplementation over multiple breeding seasons, may represent a sensitive index of environmental quality on reclaimed lands that effectively gauges avian parental strategies induced by variation in natural resource availability.

5. Conclusions

Our study showed that while reproductive success of TRES breeding on reclaimed mine lands was comparable to that of TRES breeding on the reference area, some measures of the performance of MOBL on reclaimed land were lower, and our results for MOBL additionally suggest greater annual fluctuations in ecological conditions within terrestrial systems on reclaimed land. Taken together, reproductive investment and success as well as offspring phenotypic traits of MOBL in this study indicate that additional reclamation measures, or time, may be necessary to convert tailings and waste rock areas on HVC to ecologically productive grassland systems for insectivorous birds. Such reclamation activities continue to be carried out at HVC as they move forward to achieve their primary end land use goal for the site

Table 5
Results of linear mixed models examining predictors of day 16 size and body mass of 799 nestling tree swallows raised on reclaimed mine lands and a nearby reference area (REF) over two years. Hatching mass represents the standardized mass of each nestling at day 2 after hatching, relative to the mass of all young in the brood. We set 2013 and REF study area as reference categories for parameter estimates. Brood identity was fitted as a random effect to control for common brood environment and maternal effects among siblings, and predicted all phenotypic traits (Mass: Wald Z = 6.77; Primary: Wald Z = 8.40; Head-Bill: Wald Z = 7.59; all P < 0.001).

Predictor	Body mass (g)				9th primary feather length (mm)				Head-bill length (mm)			
	df	F	P	Estimate	df	F	P	Estimate	df	F	P	Estimate
Year	1159	6.48	0.01	−13.3 ± 5.2	1157	9.00	0.003	−2.50 ± 0.8	1158	18.7	<0.001	−10.6 ± 2.4
Area	1151	15.6	<0.001	24.9 ± 6.3	1158	2.21	0.14		1153	3.52	0.06	0.21 ± 0.1
Hatching date	1151	50.1	<0.001	−0.24 ± 0.04	1158	1.97	0.16		1158	8.96	0.003	−0.05 ± 0.01
Hatching mass	1600	1.13	0.29		1597	33.2	<0.001	1.82 ± 0.2	1600	73.3	<0.001	0.17 ± 0.02
Year*Hatch Date	1159	6.73	0.01	0.08 ± 0.03					1158	19.4	<0.001	0.06 ± 0.01
Area*Hatch Date	1151	15.3	<0.001	−0.14 ± 0.04								
Area*Hatch Mass					1597	4.13	0.04	−0.46 ± 0.2				

(Freberg and Gould Gizikoff, 1999). Nonetheless, in this specific study system, we suggest that further studies of other songbirds that similarly rely on terrestrial invertebrate food sources during the breeding season could measure those traits identified here in MOBL as showing the greatest sensitivity to environmental differences between HVC and REF areas. This would allow for assessment of the generality of our results across different ecosystem components.

More generally, adaptively altering approaches to land reclamation to maximize the productivity and resilience of resulting ecosystems (Sgro et al., 2011) requires an enhanced ability to detect variation in environmental quality and assess the success of land reclamation protocols using environmentally sensitive and ecologically informative indices. Results of this study show that specific measures of avian life-history and phenotypic traits, such as timing of breeding, fledging success and offspring mass, size and feather length prior to fledging, particularly when assessed over multiple years and in the context of experimental approaches such as food supplementation, may be informative indicators of the health of reclaimed terrestrial and aquatic systems. We also suggest that it will be important for industries to collect similar data on appropriate reference areas, to provide benchmarks against which to assess performance of birds on reclaimed land. In combination, these approaches will contribute to the development of specific ecological assessment protocols for reclaimed land that are informed by life-history theory.

Acknowledgments

We thank R. Doucette for facilitating research at Highland Valley Copper, and we are grateful to B. Bogula, R. Town and all nest box trail monitors who allowed us access to boxes on the reference area. L. Schmaeder provided valuable assistance in the field. Funding was provided by Teck Highland Valley Copper Partnership, the Natural Sciences and Engineering Research Council of Canada through a Discovery Grant (238248-2010) to RDD and a Postdoctoral Research Fellowship to ELO, and a Mitacs Accelerate Fellowship to ELO. Additional financial support was provided by the University of Northern BC, Canada Foundation for Innovation (4014, 202580, 219204), BC Knowledge Development Fund (804606), and the District of Logan Lake. All agencies providing funding had no involvement in study design, the collection, analysis and interpretation of data, writing the manuscript or the decision to submit the article for publication. This study was conducted under permits from Environment Canada, and protocols were approved by the UNBC Animal Care and Use Committee on behalf of the Canadian Council on Animal Care.

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